International Journal of Heat and Mass Transfer 256 (2026) 128009

Contents lists available at ScienceDirect

International Journal of Heat and Mass Transfer

journal homepage: www.elsevier.com/locate/ijhmt

Using physics-informed machine learning for improving predictive capability of two-phase flow drift flux modeling

Zhuoran Dang [a,1,*], Yang Zhao [b,2]

[a] Department of Mechanical and Manufacturing Engineering, University of Calgary, Alberta, Canada
[b] School of Nuclear Engineering, Purdue University, West Lafayette, IN, USA

ARTICLE INFO

Keywords:
Physics-informed machine learning (PIML)
Two-phase flow models
Void fraction
Drift Flux

ABSTRACT

Two-phase flow modeling remains a challenging problem due to its inherent complexity of the phase interactions. Traditional Drift-Flux Models (DFMs) rely on empirical correlations, which are often limited to specific flow regimes or channel configurations. This work employs a Physics-Informed Machine Learning (PIML)-based approach to enhance the predictive capability of the DFM across a broader range of two-phase flow conditions. Rather than replacing the physical model, a lightweight neural network is embedded into the DFM framework to generate adaptive weighting factors for key constitutive parameters, such as the distribution parameter and drift velocity. This is realized by learning to be an expert in the applied field, especially learning the applicability of the empirical formulations based on the data under various flow conditions. The weighting factors are explicit to make the prediction traceable to its underlying physical basis. The model is trained using an experimental dataset from diverse experimental setups, including variations in flow geometry, orientation, and mass flux. Results show that the new approach enhances the DFM's prediction accuracy and stability compared to conventional drift-flux correlations. Apart from other ML application work, one highlight of this proposed PIML-based approach is its modularized structure: adjusting numbers of empirical models used does not substantially change the framework structure. It also showed that a newly defined hybrid ML framework can successfully handle small datasets containing both discrete and continuous variables. Without the need for adjustments, this approach can also be applied for other application cases where multiple empirical correlations exist, such as Critical Heat Flux or two-phase flow interfacial area modeling. The results highlighted the potential of ML-assisted physical modeling in advancing two-phase flow analysis.

1. Introduction

The Drift-Flux model (DFM) originally developed by Zuber and Findlay [1] yields good accuracy for one-dimensional two-phase flow modeling, and it can be well adapted in the Thermalhydraulics system code [2]. The model describes the two-phase flow as the mixture with the considerations of the flow distribution and relative motion between the phases. These two drift flux parameters are described using constitutive equations, and many of them have been established specifically applied for flow conditions seen in engineering applications. The drift flux analysis and model development in empirical or semi-empirical approach has generally been considered in a mature level.

One remaining issue of the DFM is that constitutive equations are limited by the flow conditions and flow channel geometries, or simply denoted as the flow setup. This is because the development and validation of these models are based on experimental observation and data analysis, which usually conducted under a similar experimental setup. Directly applying one drift flux correlation to a unvalidated range is usually considered problematic and subject to systematic error due to extrapolation. However, practical applications always require more than one model as it involves different flow channel geometries. Typically, Thermalhydraulics system code or other simulation tools will use multiple correlations with an artificial curving process to keep the prediction result continuous and smooth. This approach is still subject to model invalidity and could generate numerical instabilities.

To mitigate the effect caused by using a set of correlations in a

* Corresponding author.
E-mail address: zhuoran.dang@cnl.ca (Z. Dang).
[1] Current Affiliate: Canadian Nuclear Laboratories, Chalk River, Ontario, Canada.
[2] Current Affiliate: Argonne National Laboratory, Lemont, United States of America.

https://doi.org/10.1016/j.ijheatmasstransfer.2025.128009
Received 22 July 2025; Received in revised form 18 September 2025; Accepted 18 October 2025
Available online 24 October 2025
0017-9310/© 2025 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

continuously evolved two-phase flow, it would be ideal to use one model that covers a wide range of flow setups. However, it is impossible to describe many constitutive relationships, which are determined by several multilevel, intricate theoretical concepts [2], using a single expression with algebraic equations. Although some efforts have been made to validate the accuracy of one DFM used for another geometrical prediction [3,4], the theoretical basis used for developing the equations was not well verified. There have also been attempts to develop weighting factors (also referred to as transient functions or adjustive functions in some literature) to make one DFM applicable to other experimental setup, such as the work by Ozaki and Hibiki [5] that applied adjustable functions to Kataoka-Ishii correlation [6] for a rod bundle application. The adjustable functions are simple algebraic expressions with few physical parameters and several fitted values, which can be a temporary solution to mitigate the problem. A complete solution, although complex to achieve, is to figure out the fundamental constitutive relationships, often non-linear and high-dimensional, in terms of phase distribution and relative motions. A universal model for different flow setups should be able to describe these different relationships using one form. Developing such models poses enormous difficulties in terms of understanding of two-phase flow dynamics and data analysis for physics discovery.

Machine Learning (ML) technology provides an alternative solution approach for analyzing complex two-phase flow and developing more generalized models. Recent emerging technologies enable coupling the existing physics models and the machine learning models, known as physics-informed machine learning (PIML) or physics-guided machine learning (PGML) [7]. One approach to implementing this concept is to use the physics model as assumptions or constraints to the ML model. A typical example is designing loss functions based on the domain knowledge or physics models. Another approach is to use the ML model to enhance the physics-based model. For instance, an ML model tailored to the data structure can identify patterns related to key physical phenomena, such as detecting one or more specific flow behaviors among various possibilities.

The aim of this study is to develop a PIML-based method to improve the DFM's capability in application ranges, based on existing research efforts in DFM development. Within the method, a ML module was trained to explicitly chooses the significant physical factors corresponding to the validated correlations based on the flow setup and avoid using the correlations that are not validated in the flow setup. The newly proposed approach aims to achieve higher overall accuracy while ensuring reasonably correct and consistent selection of physical models. A significance for this work is it focuses on a data-driven approach to learn using traditional correlations with limited application ranges, learning to correctly choose the appropriately fit correlations rather than the other inappropriate ones. This approach can be applied to other similar scenarios that multiple correlation with specific application ranges, such as Critical Heat Flux or two-phase flow interfacial area modeling.

2. Drift-flux model

Drift-Flux Model (DFM) is a two-phase flow modeling approach that considers the gas and liquid phases as a mixture rather than a homogeneous phase or separate phases (Two-Fluid Model). The model accounts for void fraction distribution across the flow channel cross section and relative motion of the phases. The basic concept of the drift-flux model is drift velocity, defined as the difference between the gas velocity and volumetric flux of the two-phase mixture,

$$v_{gj} = v_g - j \tag{1}$$

Where j represents total volumetric flux of the mixture,

$$j = \alpha v_g + (1-\alpha) v_f \tag{2}$$

Where v_g and v_f are in-situ gas and liquid velocity, respectively. The one-dimensional drift-flux model can be obtained by averaging the above equation,

$$\ll v_g \gg = C_0 \langle j \rangle + \ll v_{gj} \gg \tag{3}$$

where C_0 and v_{gj} represent the distribution parameter and drift velocity, respectively. $\langle\ \rangle$ and $\ll\ \gg$ are the area-averaged quantity over the cross-sectional flow area and the void-fraction-weighted average quantity, respectively. The mathematical definition of C_0 is expressed as follows,

$$C_0 = \frac{\langle \alpha j \rangle}{\langle \alpha \rangle \langle j \rangle} \tag{4}$$

From the equation, the distribution parameter is a covariance term representing the distribution across the flow channel cross-sectional area regarding to void fraction. It captures the deviation of void distribution from uniform cross-sectional profiles. Similarly, $\ll v_{gj} \gg$ is expressed as follows,

$$\ll v_{gj} \gg = \frac{\langle \alpha v_{gj} \rangle}{\langle \alpha \rangle} \tag{5}$$

Either C_0 or $\ll v_{gj} \gg$ is hardly to be directly obtained from experimental measurements. These two variables are usually modeled using empirical correlations based on experimental data and domain knowledge. To accommodate applications with different flow channel geometries, fluid type and flow conditions, various drift-flux constitutive equations have been developed. It has been observed that among all the different aspects that affects drift flux modeling, flow channel geometry and flow regime are the key parameters that affects distribution parameters and drift velocity.

2.1. Channel geometry effect on distribution parameter

The flow channel geometry has been widely recognized to have an influence on the distribtuion parameter, C_0. For example, it has been pointed out by Ishii that the asymptotic distribution parameter, C_∞, is equal to 1.2 for circulation pipe with an aspect ratio of 1, while for rectangular channels C_∞ becomes 1.35 [8]. For vertical upward bubbly flows, bubble radial motion due to transverse forces, such as life force, will lead to non-uniform distribution across the channel cross-section (e.g. core-peaked vs. wall-peaked profile). This is known to be related to the bubble diameter. To model the non-uniformity, void fraction or ratio of volumetric flow rates (j_g/j) are also used in some studies to replace bubble diameter as they are approximately related and are relatively easy to measure.

Smaller or non-circular channels constrain bubble movement, often concentrating gas toward the center or wall, thus increasing C_0 relative to larger or round pipes [9]. Studies show that correlations developed for circular pipes are inadequate for rod-bundle sub-channels or rectangular channels, where C_0 can differ significantly [3].

2.2. Channel geometry effect on drift velocity

Some effects of channel geometry on drift velocity, V_{gj}, can be understood as the channel geometry affects the flow regime transition, thus influencing V_{gj}. For example, a well-recognized phenomenon is that there is no slug flow in large diameter pipes since a slug bubble that occupies almost all the flow channel cross-section cannot sustain stable interface [10]. In large-diameter pipes or annuli, internal recirculation patterns lead to higher V_{gj}, especially in churn regimes [11]. Besides, drift velocity is orientation dependent. In horizontal pipes, buoyancy-driven slip largely disappears, yielding $V_{gj} \sim 0$. Many horizontal correlations explicitly set drift velocity to zero based on this physics [12] (Table 1).

Table 1
Drift flux correlations used in the current study.

Literature	C_0	$\langle\langle v_{gj}\rangle\rangle$	Application range
Ishii [2]	$C_\infty - \{C_\infty - 1\}\sqrt{\frac{\rho_g}{\rho_f}}$ $C_\infty = 1.2$ *for circular pipe* $C_\infty = 1.35$ *for rectangular channel*	$\sqrt{2}\left(\frac{g\sigma\Delta\rho}{\rho_f^2}\right)^{\frac{1}{4}}(1-\langle\alpha\rangle)^{1.75}$ Bubbly $0.35\left(\frac{gD\Delta\rho}{\rho_f^2}\right)^{\frac{1}{2}}$ Slug $\sqrt{2}\left(\frac{g\sigma\Delta\rho}{\rho_f^2}\right)^{\frac{1}{4}}$ Churn-turbulent	Small and medium vertical upward pipe/ rectangular channel, $D/\sqrt{\frac{\sigma}{g\Delta\rho}} \le 40$; bubbly to churn-turbulent flow depending on use of drift velocity
Kataoka-Ishii [6]	$C_\infty - \{C_\infty - 1\}\sqrt{\frac{\rho_g}{\rho_f}}$ $C_\infty = 1.2$	$N_{\mu f} \le 2.25 \times 10^{-3}$ (*low viscous*) : $\left\langle\left\langle v^+_{g,KI}\right\rangle\right\rangle = 0.0019 D_H^{*0.809} N_\rho^{-0.157} N_{\mu f}^{-0.562}$ for $D^*_H \le 30$ $\left\langle\left\langle v^+_{g,KI}\right\rangle\right\rangle = 0.030 N_\rho^{-0.157} N_{\mu f}^{-0.562}$ for $D^*_H \ge 30$ $N_{\mu f} > 2.25 \times 10^{-3}$ (*high viscous*) : $\left\langle\left\langle v^+_{g,KI}\right\rangle\right\rangle = 0.92 N_\rho^{-0.157}$	Large diameter vertical upward pipes, and pool systems; bubbly to churn-turbulent flow
Hibiki-Tsukamoto [13]	$C_\infty - \{C_\infty - 1\}\sqrt{\frac{\rho_g}{\rho_f}}$ $C_\infty = 1.2 - 0.2\exp\left(-60.63\langle\alpha_{KI}\rangle^{2.367}\right)$ $\langle\alpha_{KI}\rangle = \frac{\left\langle j_g^+\right\rangle}{C_0\langle j^+\rangle + \left\langle\left\langle v^+_{gj,KI}\right\rangle\right\rangle}$	$\left\langle\left\langle v^+_{gj}\right\rangle\right\rangle = \sqrt{2}(1-\langle\alpha_{KI}\rangle)^{1.75}\exp\left(-60.63\langle\alpha_{KI}\rangle^{2.367}\right) + \left\langle\left\langle v^+_{gj,KI}\right\rangle\right\rangle\left(1-\exp\left(-60.63\langle\alpha_{KI}\rangle^{2.367}\right)\right)$	Vertical upward medium-to-large-size channels D 0.0508 - 0.305 m; jg 0.01-11.2m/s; jf 0-2.6 m/s; P=0.1- 4.6 MPa,
Goda et al. [14]	$(-0.0214\langle j^+\rangle + 0.772) + (0.0214\langle j^+\rangle + 0.228)\sqrt{\frac{\rho_g}{\rho_f}}$, *for* $-20m/s \leqslant \langle j^+\rangle < 0$ $(0.2\exp(0.00848(j^+ > +20) + 1.0) - 0.2exp(0.00848(j^+ > +20))\sqrt{\frac{\rho_g}{\rho_f}}$, for $\langle j^+\rangle < -20m/s$	$\sqrt{2}\left(\frac{g\sigma\Delta\rho}{\rho_f^2}\right)^{\frac{1}{4}}$	Vertical downward pipe D=16-102 mm; j=-0.45 - -24.6m/s; P=0.1-1.5 MPa
Dong-Hibiki [15]	$C_\infty = 0.900\exp\left[0.286\left(\frac{\frac{\left\langle j_g^+\right\rangle}{\langle j^+\rangle}}{0.9}\right)^{1.5}\right]$ for $\frac{\left\langle j_g^+\right\rangle}{\langle j^+\rangle} \le 0.9$ $C_\infty = -2\frac{\left\langle j_g^+\right\rangle}{\langle j^+\rangle} + 3$ for $0.9 \le \frac{\left\langle j_g^+\right\rangle}{\langle j^+\rangle} \le 1$	$\sqrt{2}\left(\frac{g\sigma\Delta\rho}{\rho_f^2}\right)^{\frac{1}{4}}$	Vertical downward pipe, D=9.53-203 mm; jg=-0.00374 - -47.2m/s; jf=-0.005 - -7m/s.
Bhagwat-Ghajar [16]	$\frac{2-\left(\frac{\rho_g}{\rho_f}\right)^2}{1+\left(\frac{Re_m}{1000}\right)^2} + \frac{\left[\left(\sqrt{1+\left(\frac{\rho_g}{\rho_f}\right)^2\frac{\cos\theta}{(1+\cos\theta)}}\right)^{(1-\alpha)}\right]^{\frac{2}{5}}}{1+\left(\frac{1000}{Re_m}\right)^2}$	$V_d = C_2C_3C_4 \times (0.35\sin\theta + 0.45\cos\theta) \times \sqrt{gD\frac{(\rho_l-\rho_g)(1-\alpha_g)}{\rho_l}}]$ $C_2 = \begin{cases}\left(\frac{0.434}{\log_{10}\left(\mu_f/0.001\right)}\right)^{0.15}, & \left(\frac{\mu_l}{0.001}\right) > 10\\ 1, & otherwise\end{cases}$ $C_3 = \begin{cases}(La/0.025)^{0.9}, & La < 0.025\\ 1, & otherwise\end{cases}$ $C_4 = \begin{cases}-1, & 0^\circ < \theta < -50^\circ \text{ and } Fr_{sg} \le 0.1\\ 1, & otherwise\end{cases}$	Pipe flow, D=0.5-305 mm; P=0.1-18.1 MPa; μ_f=10^{-4}-0.6 Pas

(*continued on next page*)

Table 1 (*continued*)

Literature	C_0	$\langle\langle v_{gj}\rangle\rangle$	Application range
Rassame-Hibiki [12]	$C_\infty - \{C_\infty - 1\}\sqrt{\frac{\rho_g}{\rho_f}}$ $C_\infty = 0.8\exp\left[0.815\left(\frac{\frac{\langle j_g^+\rangle}{\langle j^+\rangle}}{0.9}\right)^{1.5}\right]$ for $\frac{\langle j_g^+\rangle}{\langle j^+\rangle} \le 0.9$ $C_\infty = -8.08\frac{\langle j_g^+\rangle}{\langle j^+\rangle} + 9.08$ for $0.9 \le \frac{\langle j_g^+\rangle}{\langle j^+\rangle} \le 1$	0	Horizontal pipe, D=19-77.9 mm; jg=0.0253 – 47.5m/s; jf=-0.000057 – 5.97m/s.
Woldesemayat-Ghajar [17]	$\frac{j_g}{j}\left(1 + \left(\frac{j_f}{j_g}\right)^{\left(\frac{\rho_g}{\rho_f}\right)^{0.1}}\right)$	$2.9(1.22)^{\frac{P_{atm}}{P_{system}}}\left[\frac{2gD\sigma(\rho_f - \rho_g)}{\rho_f^2}\right]^{\frac{1}{4}}$	Pipe flow, D=12.7-102.26 mm
FRIGG correlation [5]	$C_\infty = 1.03$	$\langle\langle v_{gj}^+\rangle\rangle = \sqrt{2}(1 - \langle\alpha_{KI}\rangle)^{1.75}\exp\left(-1.39\langle j_g^+\rangle\right) + \langle\langle v_{gj,KI}^+\rangle\rangle\left(1 - \exp\left(-1.39\langle j_g^+\rangle\right)\right)$	8 × 8 Rod bundle, Pressure up to 8.7 MPa; m = 470–2160 kg/m^{2}s;
Hibiki-Tsukamoto [18]	$C_\infty = 1.1$	$\langle\langle v_{gj}^+\rangle\rangle = \langle\langle v_{gj,B}^+\rangle\rangle\exp\left(-60.63\langle\alpha_{KI,w}\rangle^{2.367}\right) + \langle\langle v_{gj,P}^+\rangle\rangle\left(1 - \exp\left(-60.63\langle\alpha_{KI,w}\rangle^{2.367}\right)\right)$ $\langle\langle v_{gj,B}^+\rangle\rangle$ *Ishii's bubbly correlation with D=w* $\langle\langle v_{gj,P}^+\rangle\rangle$ *Kataoka Ishii's correlation with D=w*	Rod bundle, De= 11.5 - 21.3 mm; P=0.1 - 12 MPa; m = 0.0 – 2000 kg/m^{2}s;
Abbs-Hibiki [19]	$C_\infty = \exp\left\{0.583\left(\frac{\langle j_g^+\rangle}{\langle j^+\rangle}\right)^{1.44}\right\}$ for $\frac{\langle j_g^+\rangle}{\langle j^+\rangle} \le 0.9$ $C_\infty = -3\left(\frac{\langle j_g^+\rangle}{\langle j^+\rangle}\right) + 4.35$ for $0.9 \le \frac{\langle j_g^+\rangle}{\langle j^+\rangle} \le 1$	$\langle\langle v_{gj}^+\rangle\rangle = \sqrt{2}(1 - \langle\alpha_{KI}\rangle)^{1.75}\exp\left(-1.39\langle j_g^+\rangle\right) + \langle\langle v_{gj,KI}^+\rangle\rangle\left(1 - \exp\left(-1.39\langle j_g^+\rangle\right)\right)$	Rectangular Channel, W 10-11m m, H 94-200 mm; jg=0.004 – 46.5m/s; jf= 0.315 – 4.4m/s; P=0.1-2.14MPa
Rassame-Hibiki [20]	$C_\infty - \{C_\infty - 1\}\sqrt{\frac{\rho_g}{\rho_f}}$ $C_\infty = 1.35 - 0.35\exp\left(-60.63\langle\alpha_{KI}\rangle^{2.367}\right)$ $\langle\alpha_{KI,w}\rangle = \frac{\langle j_g^+\rangle}{C_0\langle j^+\rangle + \langle\langle v_{gj,KI}^+\rangle\rangle}$	$\langle\langle v_{gj}^+\rangle\rangle = \sqrt{2}(1 - \langle\alpha_{KI}\rangle)^{1.75}\exp\left(-60.63\langle\alpha_{KI}\rangle^{2.367}\right) + \langle\langle v_{gj,KI}^+\rangle\rangle\left(1 - \exp\left(-60.63\langle\alpha_{KI}\rangle^{2.367}\right)\right)$	Rectangular channel, W 4 - 136 mm H 27 - 200 mm; jg= 0.00689 - 23.9 m/s; jf= 0 - 10.5 m/s; P =0.10 - 0.20 MPa for adiabatic condition; P = 0.42 - 5.5 MPa for boiling condition.
Wang et al. [21]	$C_0 = \frac{2}{1 + \left(\frac{Re_{tp}}{1000}\right)^2} + \frac{1.2 - 0.2\langle\alpha\rangle^4}{1 + \left(\frac{1000}{Re_{tp}}\right)^2}$	$\langle\langle v_{gj}\rangle\rangle = 1.53\left(\frac{g\sigma\Delta\rho}{\rho_f^2}\right)^{\frac{1}{4}} + 0.35\sqrt{\frac{\Delta\rho g D_H}{\rho_f}}\langle\alpha\rangle(1 - \langle\alpha\rangle)^{0.25}$	Annulus, D_{avg} = 91.5 mm; D_h = 57 mm.
Barati-Hibiki [22]	$C_\infty = 1.1 - 0.1\exp\left(-7.32\langle\alpha_{KI}\rangle^{2.17}\right)$	Hibiki and Tsukamoto's Rod Bundle $\langle\langle v_{gj}^+\rangle\rangle$ [[18]]	Annulus, D_{avg} = 22.5-108 mm; D_h=15-89 mm; jg= 0.0042 – 15.5m/s; jf= 0 – 3.32m/s;

3. PIML for drift-flux model

3.1. Motivation and objective

Traditional DFM relies on empirical correlations that are geometry-dependent and often limited to specific experimental conditions. Switching between different constitutive equations can introduce numerical instability, particularly in complex systems such as nuclear reactor cores and multiphase flow pipelines. To address these challenges, this study leverages Physics-Informed Machine Learning (PIML) to improve DFM's predicting capability for a broaden application range, i.e., across various flow conditions and geometries. It leverages the well developed drift flux correlations over the past decades and experimental databases.

The main objective for this PIML application study is to develop a ML module that can identify dominant physical parameters under certain flow setups. The dominant physical parameters were summarized and embedded in the correlation specific for the flow setup. Given a certain flow setup, a successfully developed ML module should be able to identify the dominant physical parameters associated with the correlation developed for this flow setup. As a result, using the developed ML module should give us optimal prediction results under various tested flow setups.

3.2. Physics-informed machine learning

This study leverages the concept of the PIML, whose fundamental idea is to utilize established physical laws as guides during the machine learning model's training phase, thereby ensuring that predictions and extrapolations align with established physical principles [23]. Given that the definition of PIML is broad, the approaches to implement PIML vary among studies. Physical laws can be served as reference, constraints, priors, or regularizers for the learning process, thereby integrating domain knowledge with data-driven models for enhanced accuracy and interpretability [7,23–25].

This study proposed a new form of applying physics informed machine learning, denoted as PIML-E (expert). The basic idea is to use ML model for learning to select among different physical models that describe similar physical phenomena, for example, different constitutive equations for two-phase flows in various types of flow channels. The ML model learns from physical models and experimental data to improve the model performance by selecting the most appropriate physical models for a given case scenario. Like an expert under the research topic, the trained ML model is capable of selecting valid and accurate physical models based on the input variables to establish a best-estimate model, as shown in Fig. 1. Given a specific set of flow conditions, the model dynamically assigns weights to different models by weighing more on the valid models and less/none on the invalid ones. This design improves the prediction performance compared to using a single, specifically designed correlation. Additionally, it mitigates the "black box" issue common in AI-driven systems, enabling more reliable interpretation. Understanding the targeted phenomenon can be realized by referring to the ML expert model's selection of the physics models.

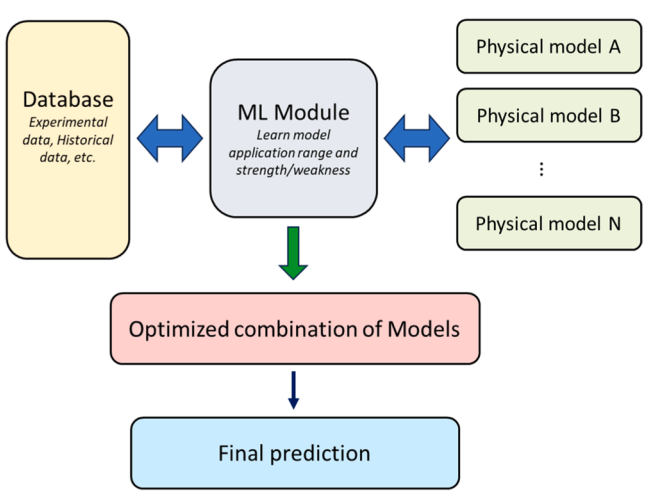

Fig. 1. Concept of machine learning module that chooses best estimate models based on data to realize interpretable, generalized predictions.

3.3. Applying PIML-E to drift flux model

The idea of the PIML-E is used for improving Drift-Flux Model's prediction capability, denoted as PIE-DF (physics-informed ensemble drift flux modeling). As depicted in Fig. 2, the PIE-DF utilizes multiple drift-flux constitutive equations based on its intended application purpose, such as developing a geometrically universal model. These equations are denoted as $\boldsymbol{C_{0i}} \ll \boldsymbol{v_{gj,i}} \gg$. The ML model is trained to become an expert in using the existing drift flux correlations, according to the flow condition and geometric configuration. As described in the "Hybrid ML module section" of Fig. 2, the ML model utilizes a hybrid architecture consisting of a multi-layer perceptron (MLP) block and an Embedding block (one Embedding layer followed with an MLP). They are placed in parallel and their outputs are concatenated with additional fully connected layers, which finally output the weights associated with the drift flux correlations. The MLP block receives continuous parameters as input, while the Embedding block is designed for discrete parameters. The continuous parameters are those that are generally distributed evenly throughout the database, including two-phase flow Reynolds numbers ($\boldsymbol{Re_k} = \frac{\rho_k j_k D}{\mu_k}$), Weber numbers ($\boldsymbol{We_k} = \frac{\rho_k j_k^2 D}{\sigma}$), Froude numbers ($\boldsymbol{Fr_k} = \frac{j_k}{\sqrt{gD}}$), viscosity number ($\boldsymbol{N_\mu} = \frac{\mu}{\sqrt{\rho\sigma La}}$, Laplace number $\boldsymbol{La} = \sqrt{\frac{\sigma}{g\Delta\rho}}$), and velocity number ($\boldsymbol{N_v} = \left(\frac{\sigma g \Delta\rho}{\rho_f^2}\right)^{1/4}$). The discrete parameters includes both categorical features (flow channel type) and physical parameters (e.g., pipe diameter, area ratio) that are represented as a limited set of distinct values in the dataset. The ML model was provided with both of the two types of parameters that related to the two-phase flow drift flux analysis with a focusing on ones related to the geometry and flow regime.

The outcome of the PIE-DF explicitly shows which drift flux constitutive equation it mainly utilizes to predict a specific flow condition using the weight vector $\boldsymbol{w_i}$. The final predicted gas velocity is,

$$\langle\langle v_g \rangle\rangle_{PIE-DF} = \sum_{i=1}^{N} w_i \left(C_{0,i} \langle j \rangle + \langle\langle v_{g,j,i} \rangle\rangle \right) \tag{6}$$

The weighting factors in PIE-DF represent the relative contribution of each empirical correlation to the final prediction. Each empirical correlation included in the PIE-DF framework is associated with a specific range of physical applicability and theoretical justification, such as particular flow regimes or channel geometries. The weighting vector generated by the PIE-DF at each prediction instance quantitatively represents the degree to which each empirical correlation is selected or combined under the given input conditions. Essentially, this mechanism functions as an "expert selection" system, allowing the model to adaptively assign higher weights to physically appropriate correlations while suppressing those that are less relevant. In this way, the weighting factors serve as interpretable indicators that directly link model predictions to the underlying physics of the problem.

The design of the weight vector $\boldsymbol{w_i}$. in the PIE-DF framework is based on the physical applicability and limitations of the underlying empirical correlations. Each empirical correlation incorporated into the model is developed and validated for specific channel geometries and flow regimes. The ground truth weighting vector is therefore constructed by

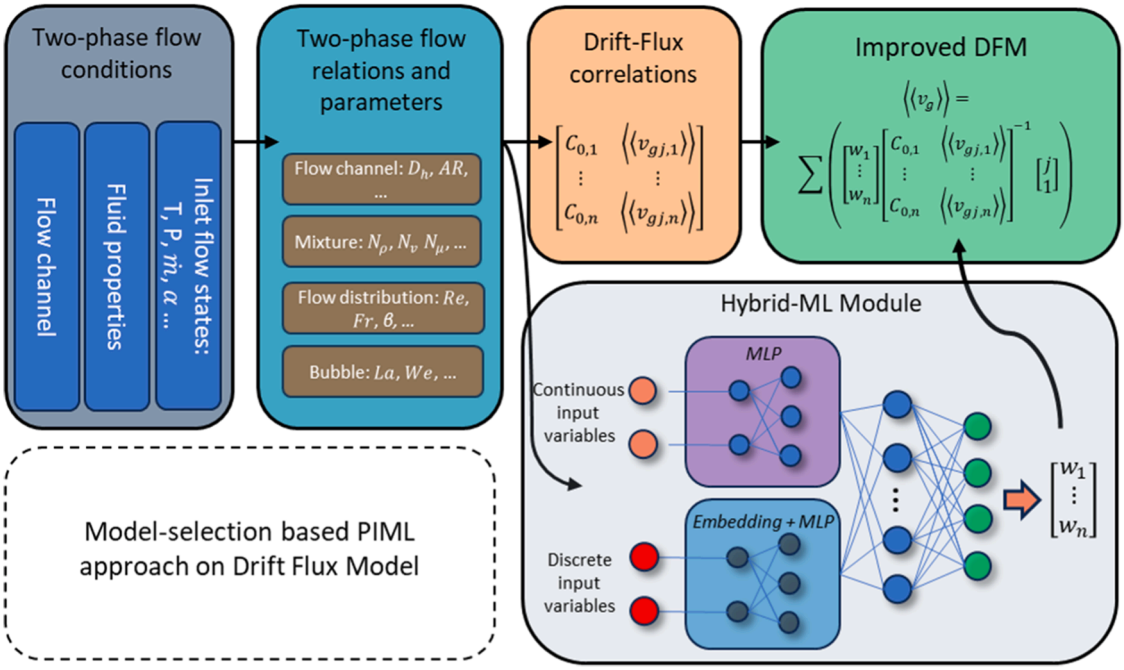

Fig. 2. Schematic of the model-selection based physics-informed learning framework to improve drift-flux model, PIE-DF.

mapping each data point to the set of physically appropriate correlations, reflecting the actual domain knowledge regarding the correlation's application range. Specifically, for a given flow scenario (e.g., vertical upward pipe), only the relevant correlations are assigned nonzero values in the weight vector, with their sum normalized to one, while all others are set to zero. In addition, a supervised loss term (e.g., cross-entropy) is introduced during model training to penalize the selection of correlations outside their validated regimes. This ensures that the learned weights are not solely driven by data fitting but are also constrained to be physically meaningful. Consequently, the weighting mechanism in PIE-DF serves as an interpretable bridge between empirical modeling and machine learning, where each predicted weighting factor has a direct physical interpretation linked to the underlying flow configuration and regime.

Using PIE-DF does not have additional complexity compared to using traditional drift flux correlations. It requires two-phase flow conditions and properties as initial input. PIE-DF uses these inputs to calculate several dimensionless parameters and relations in terms of flow setups. These parameters and relations will be fed into both the ML expert model as the ML input and the selected drift flux correlations as calculation input.

4. Model development

4.1. Database

To train and test the new model, a comprehensive set of experimental two-phase flow data was collected and summarized in Table 2. The databases used in this study come from 23 different studies and include 1392 experimental data points for air-water, steam-water, and air-kerosene two-phase flows in various types of flow channel geometries, including circular pipes (horizontal, vertical downward and upward flow conditions), rectangular channels, rod bundles, and annulus channels.

Fig. 3 shows the database distributions in terms of void fraction and flow channel type. Note that most of the data, especially non-circular flow channel data, included in this current database were originally used for developing the selected drift flux correlations. Flow conditions with void fraction larger than 0.85 are excluded for this database (mostly from horizontal circular pipes). Fig. 4 shows the range of each study's superficial velocities.

4.2. Model training

Data preprocessing was performed before data was fed into the ML expert model. Data, i.e., two-phase flow relations and parameters calculated from given two-phase flow conditions, was split into training and testing subsets in a ratio of 7:3. These subsets were normalized to their standard range separately. The training subset was further divided into a training and validation subset in a ratio of 9:1 during the training process.

Training the ML expert model uses a customized loss function that is designed to make the model informed by domain physics. The approach to establish this loss function is to consider both the overall prediction accuracy – namely, the correctly predicted weight vector - and the choice of model that a given flow condition is appliable for. This type of loss function is called the supervised loss function in this study, $\boldsymbol{L_s}$, as it provides evaluation on the choice of model. The model prediction accuracy and choice of model are quantified using root mean squared error (RMSE) and cross entropy, respectively. The mathematical form of $\boldsymbol{L_s}$ is,

$$L_s = -\frac{\xi}{N}\sum_{i=1}^{N}\sum_{j=1}^{N} w_i \log(\widehat{w}_j)\langle\langle v_g\rangle\rangle + \sqrt{\left(\langle\langle v_g\rangle\rangle - \langle\langle v_g\rangle\rangle_{M\,LmDFM}\right)^2} \quad (7)$$

where $\langle\langle v_g\rangle\rangle$, w, and $\widehat{w}$ are the bubble velocity from experiment as ground truth, predicted weight vector and true weight vector, respectively. The true weight vector is determined by the predicted flow condition and the model application range. For example, all the models for rectangular channel (Abbs-Hibiki [19] and Rassame-Hibiki [20]) are marked with equal values, with a sum equal to one in a case of rectangular two-phase flow, while the rest of the models are marked with zero. N is the number of classes, i.e. number of drift flux correlations in this study. ξ is an adjustable factor that tunes the model's attention to the correct choice of model or the overall model prediction. The setup of ξ should make sure that the values generated from RMSE and cross entropy are comparable to prevent one objective from being overly

Table 2
Summary of Databases Used in This Study.

Study	Fluids	Channel Type	Channel Dimension [mm]	Number of Data	j_g (m/s)	j_f (m/s)	System pressure (MPa)	Correlation developed based on this data
Kim (1999) [26]	Air-water	Rectangular	10 × 200	16	0.0400–0.854	0.315–4.40	0.1	Abbs-Hibiki [19]; Rassame-Hibiki [20]
Sun (2001) [27]	Air-water	Rectangular	10 × 200	26	0.321–1.84	0.315–2.84	0.1	Abbs-Hibiki [19]; Rassame-Hibiki [20]
Liu (2008) [28]	Air-water	Rectangular	10 × 200	32	0.00400–1.48	0.515–2.32	0.1	Abbs-Hibiki [19]; Rassame-Hibiki [20]
Marchaterre (1956) [29]	Steam-water	Rectangular	11 × 94	30	2.92–26.5	1.46–2.32	0.79-4.23	Abbs-Hibiki [19]; Rassame-Hibiki [20]
Zhang et al. (2022) [30]	Air-water	Rectangular	4 × 66	24	0.094-3.45	0.5-2.0	0.1	Rassame-Hibiki [20]
Chalgeri and Jeong (2019) [31]	Air-water	Rectangular	2.35 × 66.6	149	0.02–7.08	0.3–3.5	0.1	
Yang et al. (2012) [32]	Air-water	Rod Bundle	8 × 8	57	0.065-7.53	0.088-1.4	0.1, 0.3	Hibiki-Tsukamoto [18]
NUPEC (1989) [33]	Steam-water	Rod Bundle	8 × 8; $D_0 = 12.3$; $P_0 = 16.2$; $D_h = 14.9$; $w = 132.5$	51	0.295–31.1	0.275–2.55	1.0–8.6	Hibiki-Tsukamoto [18]
ORNL (1983) [34]	Steam-water	Rod Bundle	8 × 8; $D_0 = 9.5$; $P_0 = 12.7$; $D_h = 12.1$; $w = 104$	54	0.03–3.56	0.00–0.11	3.9–8.1	Hibiki-Tsukamoto [18]
JAERI (1994) [35]	Steam-water	Rod Bundle	32; $D_0 = 9.5$; $P_0 = 12.6$; $D_h = 11.8$; $w = 75.6$	72	0.07–4.99	0.000–0.126	3–12	Hibiki-Tsukamoto [18]
Sadatomi et al. (1982) [36]	Air-water	Annulus	$D_i = 15$; $D_o = 30$	27	0.082–4.3	0.099–2.0		Barati-Hibiki [22]
Wang et al. (2017) [37]	Air-water	Annulus	$D_i = 38,71,89$; $D_o = 127$	29	0.008–0.20	0.017-0.1	0.1	Barati-Hibiki [22]
Ozar et al. (2008) [38]	Air-water	Annulus	$D_i = 19.1$; $D_o = 38.1$	21	0.04–3.43	0.240–3.32	0.1	Barati-Hibiki [22]
Caetano et al. (1992) [39]	Air-water	Annulus	$D_i = 42.2$; $D_o = 76.2$	66	0.0125–14.69	0.002–3.05	0.2-0.43	Barati-Hibiki [22]
Caetano et al. (1992) [39]	Air-Kerosene	Annulus	$D_i = 42.2$; $D_o = 76.2$	72	0.029–10.7	0.03–2.4	0.32-0.4	Barati-Hibiki [22]
Dang et al. (2018-19) [40,41]	Air-water	Downward	$D_h = 101.3, 152.4$	51	0.06-9.22	0.19-2.97	0.1	
Usui (1989) [42]	Air-water	Downward	$D_h = 16, 24$	123	0.021-13.378	0.091-1.43	0.1	Dong-Hibiki [15]
Kong (2018) [43]	Air-water	Horizontal	$D_h = 38.1$	62	0.067-3.0	1.0-6.0	0.1	Rassame-Hibiki [12]
Dang et al. (2018-19) [40,41]	Air-water	Horizontal	$D_h = 152.4$	60	0.06-9.22	0.19-2.97	0.1	
Abdul-Majeed (1996) [44]	Air-Kerosene	Horizontal	$D_h = 50.8$	71	0.196-48.908	0-1.825	0.8	Rassame-Hibiki [12]
Dang et al. (2016-17) [45,46]	Air-water	Upward	$D_h = 25.4$	153	0.03-30	0.2-2.7	0.1	
Dang et al. (2021) [9]	Air-water	Upward	$D_h = 12.7$	69	0.078-2.301	0.3-2.0	0.1	
Wang et al. (2021) [47]	Air-water	Upward	$D_h = 25.4$	72	0.1-5.0	0.5-2.0	0.1	

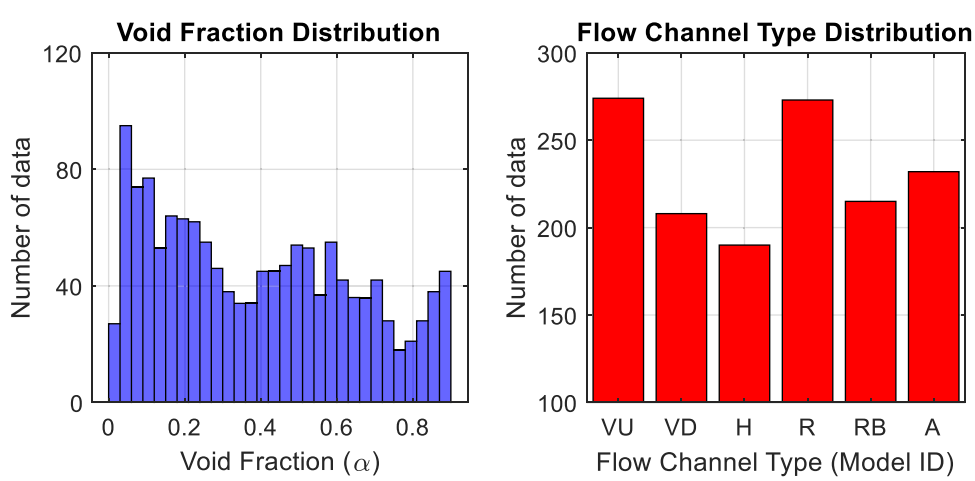

Fig. 3. Distribution of the experimental database in terms of void fraction and flow channel type.

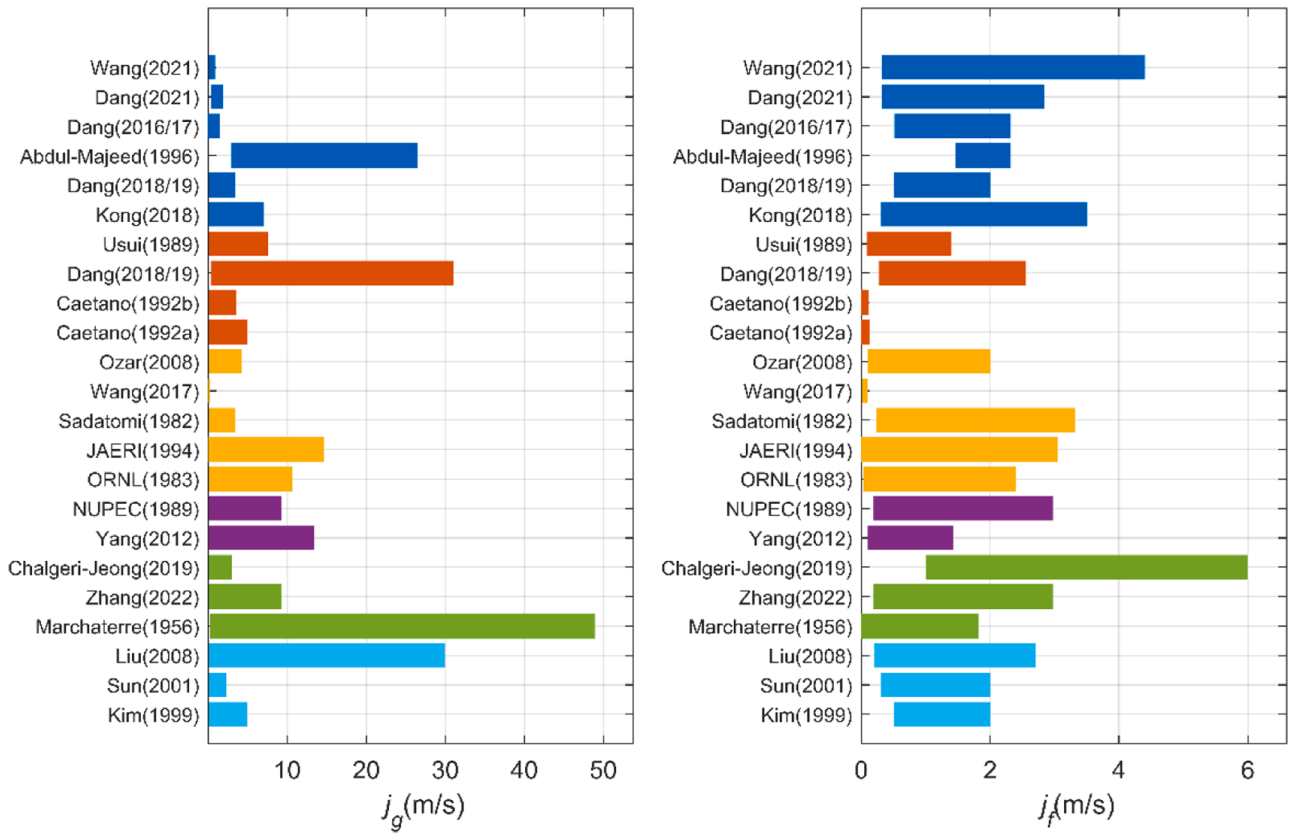

Fig. 4. Ranges of (a) gas superficial velocity and (b) liquid superficial velocity for all studies in the database. Bar color specifies the flow channel types.

emphasized during model training.

The supervised loss function evaluates the model based on not only the model prediction accuracy, but also more importantly the applicability of the ML expert model's selection on correlations. It is acknowledged that the empirical correlations used in this study were originally developed for specific flow geometries and conditions. To prevent misleading associations generated from mismatched database-correlation, this loss function serves to properly guide the ML expert model to learn to use the correct correlation that applies only within or near its validated range during the model training. It enables the model to adhere to physical constraints while still optimizing predictive performance, mitigating the risk of mismatched correlation-data pairings.

4.3. Model evaluation

Model evaluation uses the following metrics:

- Absolute Relative Error

$$\mathrm{ARE} = \left|\frac{\langle\langle v_g\rangle\rangle - \langle\langle v_g\rangle\rangle_{\mathrm{PIE-DF}}}{\langle\langle v_g\rangle\rangle}\right| \tag{8}$$

- Mean Relative Difference (MRD)

$$\mathrm{MRD} = \frac{1}{N}\sum_{i=1}^{N}\frac{\langle\langle v_g\rangle\rangle - \langle\langle v_g\rangle\rangle_{\mathrm{PIE-DF}}}{\langle\langle v_g\rangle\rangle} \tag{9}$$

- Standard Deviation of Relative Difference (SDRD)

$$\mathrm{SDRD} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left[\frac{\langle\langle v_g\rangle\rangle - \langle\langle v_g\rangle\rangle_{\mathrm{PIE-DF}}}{\langle\langle v_g\rangle\rangle} - \mathrm{MRD}\right]^2} \tag{10}$$

- Mean Absolute Error (MAE)

$$\mathrm{MAE} = \frac{1}{N}\sum_{i=1}^{N}\left|\langle\langle v_g\rangle\rangle - \langle\langle v_g\rangle\rangle_{\mathrm{PIE-DF}}\right| \tag{11}$$

- Root Mean Square Error (RMSE)

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left[\langle\langle v_g\rangle\rangle - \langle\langle v_g\rangle\rangle_{\mathrm{PIE-DF}}\right]^2} \tag{12}$$

- R^2 score (coefficient of determination)

$$\mathrm{R}^2 = 1 - \frac{\frac{1}{N}\sum_{i=1}^{N}\left[\langle\langle v_g\rangle\rangle - \langle\langle v_g\rangle\rangle_{\mathrm{PIE-DF}}\right]^2}{\frac{1}{N}\sum_{i=1}^{N}\left[\frac{1}{N}\sum_{i=1}^{N}\langle\langle v_g\rangle\rangle - \langle\langle v_g\rangle\rangle_{\mathrm{PIE-DF}}\right]^2} \tag{13}$$

Where $\boldsymbol{N}$ and $\boldsymbol{i}$ are the total number of experimental data points in the test subsets and $\boldsymbol{i}$-th data point, respectively.

4.4. Model performance

The performances of PIE-DF and all the traditional drift-flux correlations were evaluated using the test subset, which contained a total of 418 data points. The overall performance of the PIE-DF, as well as other drift-flux correlations, is presented in Table 3. The PIE-DF achieves lower prediction errors (e.g., RE, RMSE, and MAE) and higher coefficients of determination ($\boldsymbol{R}^2$) compared to other drift-flux correlations. To further demonstrate model performance, a cumulative distribution plot for all these models is provided in Fig. 5. It illustrates

Table 3
Summary of Model performance on the test data.

Model	MRD [-]	SDRD [-]	MAE [m/s]	RMSE [m/s]	R^2 [-]
Ishii	0.0542	0.5024	1.2653	3.3809	0.9292
Kataoka-Ishii	0.0982	0.5617	1.2865	3.3875	0.9289
Hibiki-Tsukamoto	0.0474	0.4848	1.1988	3.3546	0.9303
Dong-Hibiki	-0.0103	0.4244	1.3590	4.1856	0.8914
Goda et al.	-0.0153	0.4151	1.4450	4.4876	0.8752
Rassame-Hibiki	-0.0946	0.3943	1.2304	2.6798	0.9555
Bhagwat-Ghajar	0.0195	0.5250	1.6174	4.8744	0.8528
Woldesemayat-Ghajar	-0.1364	0.4601	1.5481	3.8066	0.9102
Rassame-Hibiki	0.0542	0.5024	1.2653	3.3809	0.9292
Abbs-Hibiki	0.0982	0.5617	1.2865	3.3875	0.9289
FRIGG correlation	0.0474	0.4848	1.1988	3.3546	0.9303
Hibiki-Tsukamoto	-0.0103	0.4244	1.3590	4.1856	0.8914
Wang et al.	-0.0153	0.4151	1.4450	4.4876	0.8752
Barati-Hibiki	-0.0946	0.3943	1.2304	2.6798	0.9555
PIE-DF	0.0288	0.5402	0.9689	1.9412	0.9759

the fraction of total predictions (vertical axis) that fall within a given absolute relative error threshold (horizontal axis). In Fig. 5, PIE-DF consistently exhibits better cumulative accuracy than traditional correlations for nearly the entire range of relative errors. This indicates that the PIE-DF is capable of finding optimal weights among the given drift-flux correlations for almost all cases in the test dataset.

A detailed performance comparison for predicting flows under different flow channel types is provided in Table 4. The accuracy of traditional drift flux correlations strongly depends on the specific flow channel conditions under which they were developed. They usually perform well in their intended application domain, but their accuracy degrades for cases outside this domain. On the contrary, PIE-DF shows a relatively uniform prediction performance, with AREs for each flow channel condition below the average ARE achieved by traditional drift flux correlations. At the same time, PIE-DF's overall ARE is the best among all the models. This result demonstrates PIE-DF's strength in effectively generalizing across diverse scenarios. Please note that the comparison of individual drift-flux correlations outside their validated ranges is not intended to be definitive. These results are presented solely

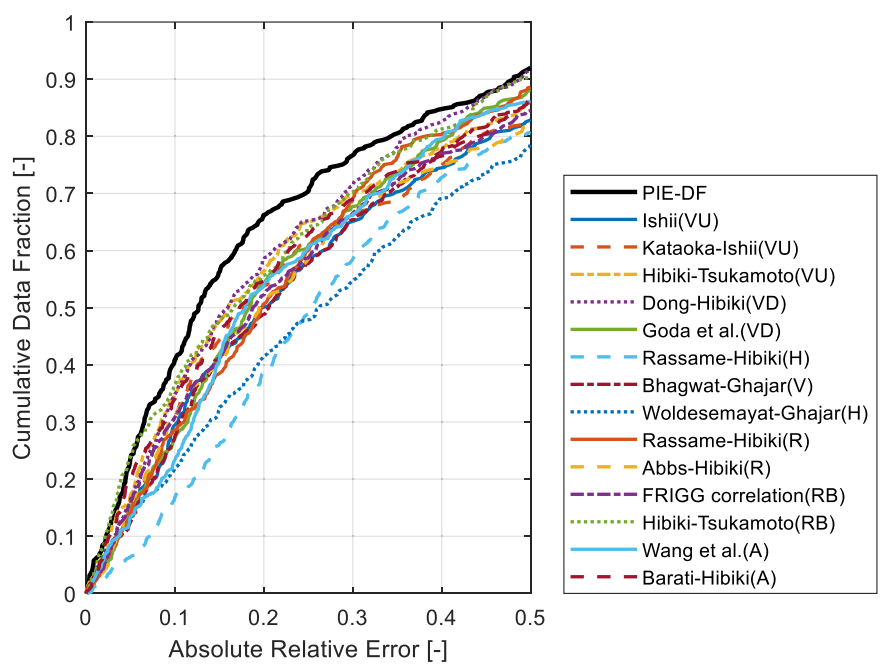

Fig. 5. Cumulative distribution plot of absolute relative errors.

Table 4
Summary of Model's ARE on each flow channel type.

Model	Vertical Upward	Vertical Downward	Horizontal	Rectangular	Rod Bundle	Annulus	Overall
Ishii	0.0808	0.6226	0.4241	0.2388	0.2840	0.2929	0.3061
Kataoka-Ishii	0.0774	0.7233	0.4573	0.2376	0.2293	0.2766	0.3134
Hibiki-Tsukamoto	0.0583	0.6413	0.3342	0.2274	0.2094	0.2705	0.2746
Dong-Hibiki	0.0850	0.5401	0.2535	0.2657	0.1879	0.2085	0.2476
Goda et al.	0.1130	0.4447	0.3728	0.2676	0.2204	0.2449	0.2667
Rassame-Hibiki	0.2138	0.4739	0.1867	0.2742	0.3599	0.4033	0.3150
Bhagwat-Ghajar	0.0839	0.5800	0.3997	0.2923	0.2549	0.2587	0.2973
Woldesemayat-Ghajar	0.2364	0.5406	0.3579	0.3937	0.2321	0.2292	0.3271
Rassame-Hibiki	0.1208	0.7672	0.3657	0.1753	0.2531	0.1869	0.2924
Abbs-Hibiki	0.1355	0.8019	0.3740	0.1797	0.1895	0.2808	0.3083
FRIGG correlation	0.0726	0.5455	0.3232	0.2918	0.2323	0.3246	0.2870
Hibiki-Tsukamoto	0.0437	0.5903	0.3669	0.2455	0.2203	0.1859	0.2597
Wang et al.	0.1173	0.8069	0.4705	0.2496	0.2550	0.1375	0.3185
Barati-Hibiki	0.0611	0.5404	0.3022	0.2715	0.2113	0.3035	0.2703
PIE-DF	0.0587(2nd)	0.4969(3rd)	0.1765(1st)	0.1788(2nd)	0.1976(3rd)	0.2071(4th)	0.2148(1st)

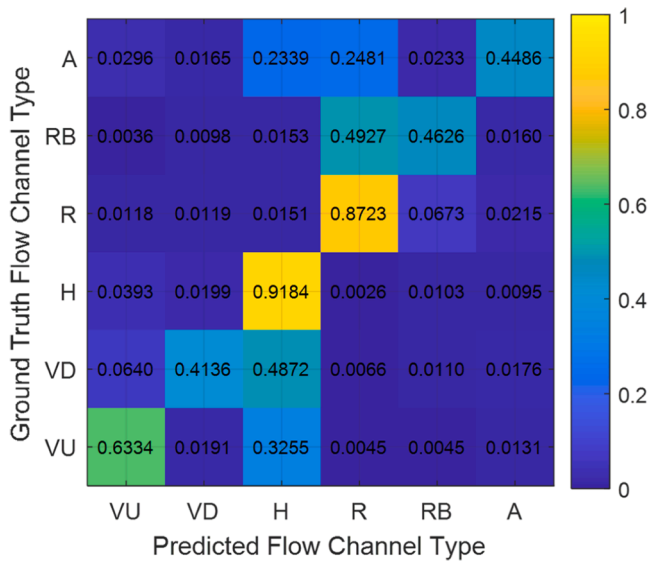

Fig. 6. True and predicted weighting vector w for each empirical correlation by PIE-DF grouped according to the flow channel type.

as reference values to demonstrate that the proposed model achieves comparable accuracy within the valid domains of the respective correlations.

The mechanism behind PIE-DF's interpretability is clarified through an analysis of model-selection behaviors. Fig. 6 shows the average weight distributions assigned by PIE-DF across different flow channel conditions, where each row represents a ground-truth flow channel type, and each cell indicates the mean weight assigned to predicted flow channel types for samples belonging to that ground-truth category. In this study, the PIE-DF assigns higher weights to the correlations that are physically appropriate for the given flow configuration. This pattern is consistent across the database and aligns well with established physical understanding. For the current test subset, it shows a tendency to favor correlations developed for horizontal and rectangular channels. This preference can be attributed to the strong performance of the empirical correlations developed for horizontal and rectangular channels in this study, as seen in Table 4. The overall accuracies of the correlations for these two types of flow channels are indeed the best among correlations. These results confirm that the weighting mechanism reflects the underlying physics, as designed.

Fig. 7 show detailed predicted gas velocity against experimental data. Traditional drift-flux correlations often display systematic prediction biases and high scatter due to their inherent geometrical dependencies. In contrast, the prediction of PIE-DF provides a closer align with the ideal prediction line and include more data points with the error margins.

5. Towards improving PIE-DF's performance: PIE-DF-E

PIE-DF is not a surrogate model that simply replaces traditional physical correlations with a data-driven black box. Instead, it acts as a physics-informed expert system that adaptively integrates and weighs multiple well-established empirical correlations based on the input flow conditions. This enables the framework to preserve the interpretability and physical consistency of conventional models while leveraging the flexibility and pattern recognition capabilities of machine learning.

To further enhance the accuracy of the PIE-DF, an error estimation function was introduced, based on the ML model was added, leading to the development of PIE-DF-E (PIE-DF-Extended). This model builds upon the existing PIE-DF framework by incorporating an additional error compensation function, which refines the predicted gas velocity by correcting systematic biases learned from the data. In this case, the drift flux model calculation is updated into,

$$\langle\langle v_g\rangle\rangle_{\mathrm{PIE-DF}-e} = \sum_{i=1}^{N} w_i\left(C_{0,i}\langle j\rangle + \langle\langle v_{gj,i}\rangle\rangle\right) + err \tag{14}$$

Where err is the error estimation value. This value comes together with the predicted weight vector as the ML output. Refering to Fig. 2, now the output of the ML becomes,

$$[w_1,\ \ldots, w_n, err] \tag{15}$$

The rest of the framework structure remains unchanged. The loss function used to train the ML model is changed into the following form,

$$\begin{aligned} L_{\mathrm{PIE-DF}-e} &= L_S + \sqrt{\left(\langle\langle v_g\rangle\rangle - \langle\langle v_g\rangle\rangle_{\mathrm{PIE-DF}-e}\right)^2} \\ &= -\frac{\xi}{N}\sum_{i=1}^{N}\sum_{j=1}^{N} w_i \log(\widehat{w}_j)\langle\langle v_g\rangle\rangle + \xi_E\sqrt{\left(\langle\langle v_g\rangle\rangle - \langle\langle v_g\rangle\rangle_{\mathrm{PIE-DF}}\right)^2} \\ &+ \sqrt{\left(\langle\langle v_g\rangle\rangle - \langle\langle v_g\rangle\rangle_{\mathrm{PIE-DF}-e}\right)^2} \end{aligned} \tag{16}$$

This loss function includes the original supervised loss function in terms of $\langle\langle v_g\rangle\rangle_{\mathrm{PIE-DF}}$ and a loss function regarding $\langle\langle v_g\rangle\rangle_{\mathrm{PIE-DF}-e}$. In other words, the loss function guides the ML model to pay attention on not only the overall performance with the error correction, but also the choice of correlation and the accuracy of the prediction without the error correction. The latter aims to preserve the prediction ability of the PIE-DF which is proved to be learned from physics.

The improved performance of PIE-DF-E is evident in Table 5, which presents all evaluation metrics used in Table 3 and Table 4. It can be seen that PIE-DF-E achieves a significant reduction in prediction errors across all key evaluation metrics. A more detailed breakdown of performance across different flow channel types (also in Table 5) further reinforces the advantages of PIE-DF-E. Across all six flow geometries, PIE-DF-E consistently exhibits lower RMSE values compared to both the original PIE-DF and traditional correlations. Fig. 8 presents cumulative error distributions, where PIE-DF-E achieves a consistently lower absolute relative error across the entire dataset compared to PIE-DF in Fig. 5., Fig. 9 provides direct comparisons of predicted versus experimental gas velocity and void fraction, which are plotted in the same manner as those plots in Fig. 7. The prediction results by PIE-DF-E reveal a tighter clustering around the ideal prediction line.

Fig. 10 shows PIE-DF-E's average predicted weight distribution for different flow channel types, plotted in the same manner as Fig. 6. With both models being trained under the concept of supervised loss, PIE-DF-E generally follows the weight distribution produced by PIE-DF. The weight distribution from PIE-DF-E shows greater diagonal dominance and slightly improved correlation selection. This pattern is consistent with physical expectations and demonstrates that the model is able to correctly identify and prioritize the underlying physical mechanisms associated with each flow configuration. The introduction of the error estimator can help transfer the attention of the ML model from the overall accuracy to model choice.

Fig. 11 provides the mean error estimation value, *err*, as a function of void fraction (left) and flow channel type (right). the mean *err* results indicate that at low void fraction, the DFMs slightly underestimate gas velocity, thus, a higher compensation is needed. Whereas at high void fraction, DFMs overestimate gas velocity, which means a reduction is needed. In terms of flow channel type, the rectangular channel type requires nearly 30% additional compensation, while other types require 5%-15% reduction. This aligns with observed traditional DFM model performance in Table 4.

6. Conclusion

This study introduced a data-driven framework, PIE-DF (Physics-Informed Ensemble for Drift-Flux modeling), which employs the concept

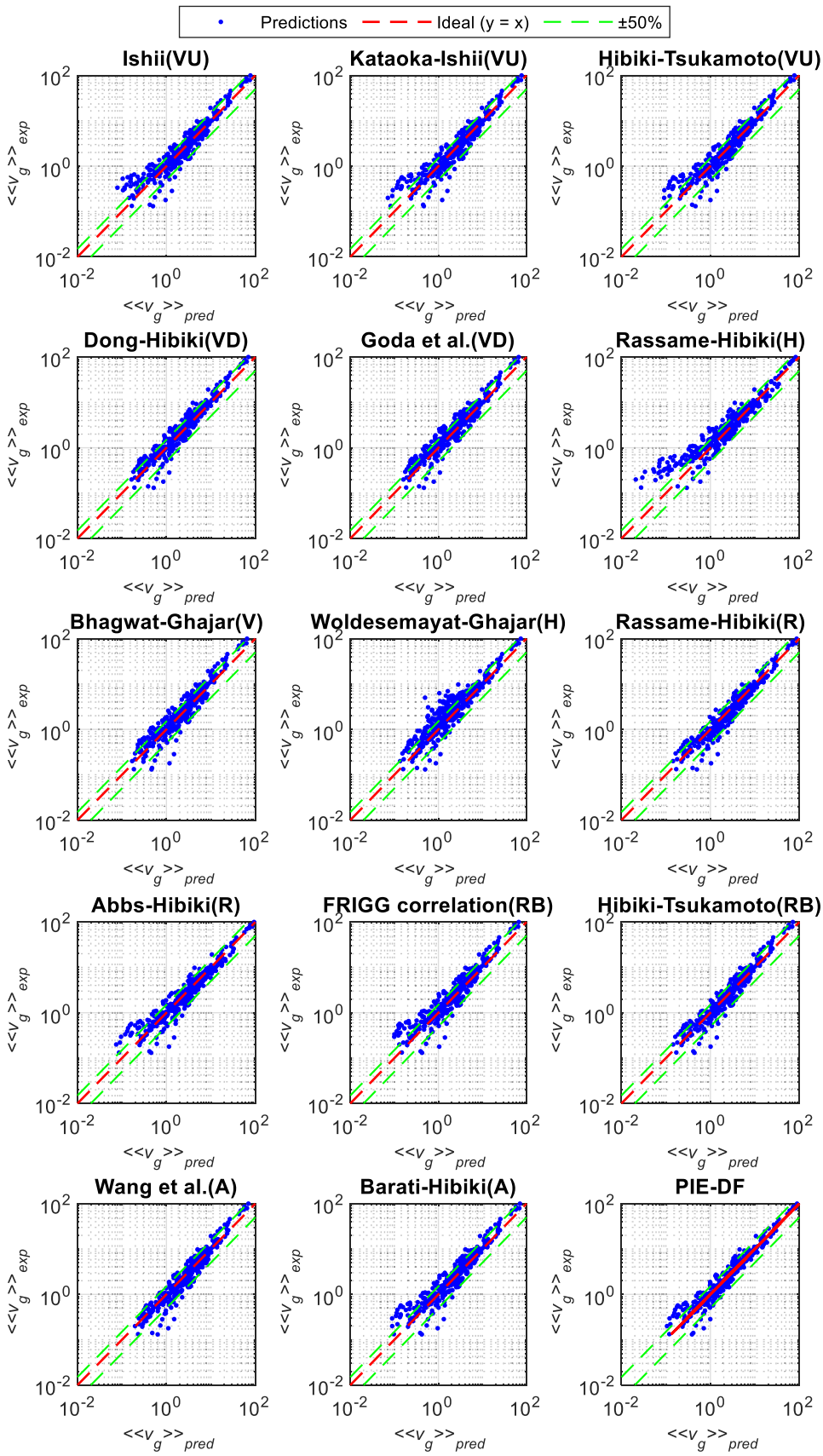

Fig. 7. Comparison of predicted versus experimental gas velocity across various empirical correlations and PIE-DF. (Model application geometry: VU: vertical upward circular pipe; VD: vertical downward circular pipe; H: horizontal pipe; V: vertical circular pipe; RB: rod bundle; R: rectangular; A: annulus.).

Table 5
Summary of PIE-DF-E's performance on the test data.

MRD [-]	SDRD [-]	MAE [m/s]	RMSE [m/s]	R^2 [-]	
0.0179	0.3492	0.4001	0.7530	0.9965	
ARE under different flow channel types					
Vertical Upward	Vertical Downward	Horizontal	Rectangular	Rod Bundle	Annulus
0.0514	0.3734	0.1689	0.1179	0.1706	0.2282

of physics-informed machine learning (PIML) to improve the capability of drift flux model across a broader range of two-phase flow conditions. Unlike simply predicting and replacing the traditional correlations, PIE-DF aims to establish a ML-based expert model that dynamically ensembles multiple existing correlations based on their known applicability and predictive performance. This ML expert model is designed as a hybrid neural network architecture that integrates multi-layer perceptrons (MLPs) with embedding layers that allows to efficiently process both continuous variables (e.g., flow conditions) and discrete descriptors (e.g., channel geometry types).

The ML expert model was trained using a dataset of 968 experimental data points, and its ability on managing multiple drift-flux correlations was evaluated with another 418 data points using multiple metrics including absolute relative error (ARE), root mean squared error (RMSE), and coefficient of determination ($\boldsymbol{R}^2$). The evaluation results demonstrated that PIE-DF provides better overall accuracy than traditional drift-flux models across a diverse flow range, while also ensuring physical consistency.

To further enhance prediction accuracy, PIE-DF-E was developed by introducing an error estimation factor into the existing framework. This extended model improved predictive accuracy by mitigating the model biases in drift-flux correlation predictions with the proof of lower ARE and MAE values across all examined flow geometries. It also shows greater robustness as it also has low standard deviation. Compared to prior work, a distinguishing feature of PIE-DF-E is that it incorporates both expert-guided model selection (i.e., optimal weighting of physical correlations) and bias correction in an integrated framework, rather than relying solely on learned residuals.

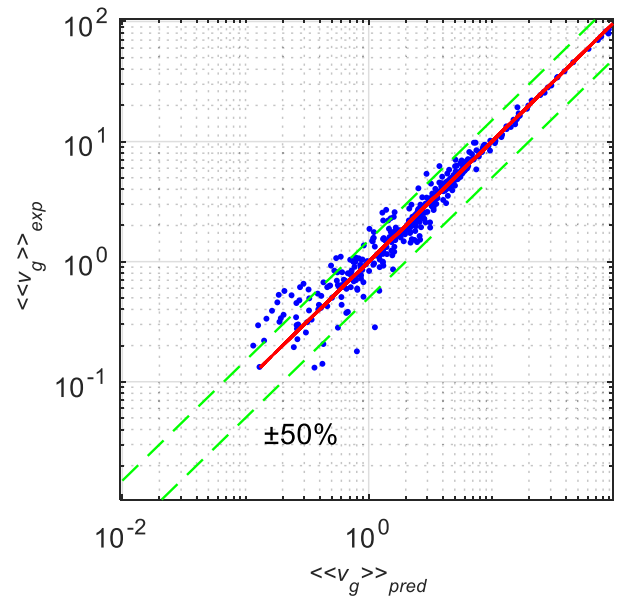

Fig. 9. Comparison of predicted vs experimental gas velocity by PIE-DF-E.

This study provides additional significance to scientific ML applications. Firstly, unlike other studies that use typical form of ML model, a hybrid ML architecture was proposed and it can be successfully trained based on around only 1000 data points. The data features in engineering applications, such as pipe diameter, appear to be continuous variables but are practically discrete since it is a fixed value for one experimental study. This hybrid ML architecture can be an example to other studies using data-driven approaches to solve engineering problems that contain both discrete and continuous features. Secondly, the proposed PIE-DF framework introduces a machine learning-based expert system that adaptively selects and combines empirical correlations to produce best-estimate predictions. It enhances accuracy while preserving the physical consistency of established models.

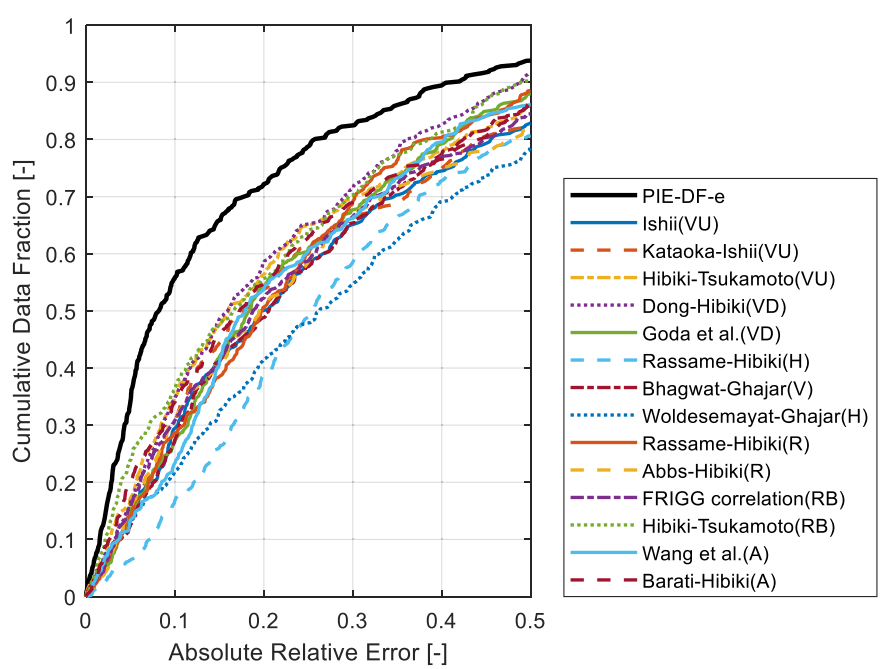

Fig. 8. Cumulative distribution plots of absolute relative errors.

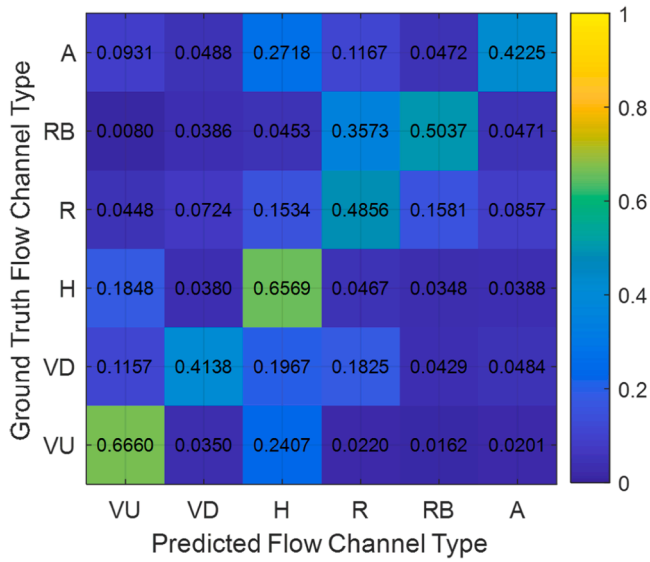

Fig. 10. True and predicted weighting vector w for each empirical correlation by PIE-DF-E grouped according to the flow channel type.

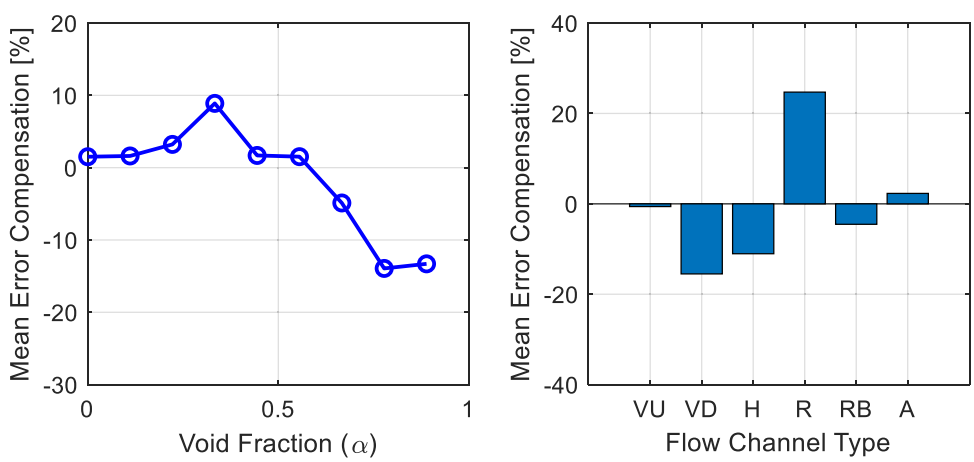

Fig. 11. Error estimator's compensation percentage (of PIE-DF gas velocity value) in terms of a) void fraction; b) flow channel type.

6.1. Limitations and future work

The current models, PIE-DF and PIE-DF-E, were trained on a dataset with limited size. its performance is constrained by the available experimental data. Specifically, the current developed model is applicable for two-phase flow from bubbly flow up to churn flow, and annular and beyond annular flows are not covered. While the ML model in the current approach shows satisfyingly good generalization capabilities, additional efforts in data expansion could further enhance predictive robustness. Additionally, hyperparameter optimization and architecture refinement could improve computational efficiency and reduce potential overfitting.

The newly proposed PIML method, PIML-E is conceptually similar to the curving methods commonly employed in thermal-hydraulics (TH) system codes to handle multiple empirical correlations with limited ranges of applicability. By adopting a data-driven approach, the present method gives potential improvements in numerical stability and smoothness. However, a direct quantitative comparison between PIE-DF and conventional curving methods was not conducted here. Such an evaluation will be necessary to fully assess the advantages and potential limitations of the proposed approach.

Future research would explore expanding the database to incorporate a wider range of flow conditions, including extreme cases and transient phenomena, and investigate the applicability of alternative physics-informed machine learning techniques to further refine correlation selection. Additionally, the framework developed in this study could be extended beyond drift-flux modeling to other two-phase flow constitutive relationships, such as critical heat flux (CHF) modeling and interfacial area transport models.

CRediT authorship contribution statement

Zhuoran Dang: Writing – original draft, Software, Resources, Methodology, Investigation, Formal analysis, Conceptualization. **Yang Zhao:** Writing – review & editing, Validation, Methodology.

Declaration of competing interest

July 21, 2025

We wish to confirm that there are no known conflicts of interest associated with this publication and there has been no significant financial support for this work that could have influenced its outcome.

We confirm that the manuscript has been read and approved by all named authors and that there are no other persons who satisfied the criteria for authorship but are not listed. We further confirm that the order of authors listed in the manuscript has been approved by all of us.

We confirm that we have given due consideration to the protection of intellectual property associated with this work and that there are no impediments to publication, including the timing of publication, with respect to intellectual property. In so doing we confirm that we have followed the regulations of our institutions concerning intellectual property.

We understand that the Corresponding Author is the sole contact for the editorial process (including Editorial Manager and direct communications with the office). He is responsible for communicating with the other authors about progress, submissions of revisions and final approval of proofs. We confirm that we have provided a current, correct email address which is accessible by the Corresponding Author.

Appendix A

Table 6 provided a summary of key information on PIE-DF model establishment. Besides, the complete code and trained models can be found in https://github.com/zhuorandang/PIE-DF.

Table 6
Summary of PIE-DF structure and training setup.

Parameter/Feature	Description	Value/Type
Model Framework	Package	Keras (TensorFlow backend)
Learning Rate	Optimization learning rate	5E-05
Activation Function	Activation for hidden layers	tanh
Batch Size	Number of samples per gradient update	32
Epochs	Number of complete training passes	2000
Continuous Inputs MLP	Hidden layer sizes for continuous inputs branch	[32, 32, 32]
Embedding Output Dimension	Dimension for discrete input embeddings	6
Discrete Inputs MLP	Hidden layer sizes for discrete inputs branch	[64, 64]
Optimizer	Optimization algorithm	Adam
Normalization	Batch normalization after each hidden layer	Enabled
Early stopping	ML model training to prevent overfitting	Patient=100
Model choice loss factor	ξ	0.3
Error Estimation	Secondary output to estimate prediction error	Enabled via custom sigmoid
Error Estimation Loss Factor	ξ_E	3
Random Seeds	Random seed that fixes the random variable to control the ML training and prediction outputs	6, 73

Data availability

Data will be made available on request.

References

[1] N. Zuber, J.A. Findlay, Average volumetric concentration in two-phase flow systems, J. Heat. Transf. 87 (4) (1965) 453–468.
[2] M. Ishii, T. Hibiki, Thermo-Fluid Dynamics of Two-Phase Flow, Springer, 2010.
[3] T. Hibiki, One-dimensional drift-flux correlations for two-phase flow in medium-size channels, Exp. Comput. Multiph. Flow. 1 (2) (2019) 85–100.
[4] A. Zolfaghari, M. Behzadi, A. Minuchehr, M.M. Mirzaee, A comparative survey on the boiling flow in different geometries using the drift-flux model, Progr. Nucl. Energy 124 (2020) 103314.
[5] T. Ozaki, T. Hibiki, Drift-flux model for rod bundle geometry, Progr. Nucl. Energy 83 (2015) 229–247.
[6] I. Kataoka, M. Ishii, Drift flux model for large diameter pipe and new correlation for pool void fraction, Int. J. Heat. Mass Transf. 30 (9) (1987) 1927–1939.
[7] G.E. Karniadakis, I.G. Kevrekidis, L. Lu, P. Perdikaris, S. Wang, L. Yang, Physics-informed machine learning, Nat. Rev. Phys. 3 (6) (2021) 422–440.
[8] M. Ishii, One-dimensional Drift-Flux Model and Constitutive Equations for Relative Motion between Phases in Various Two-Phase Flow Regimes (No. ANL-77-47), Argonne National Lab., Ill, USA), 1977.
[9] Z. Dang, G. Wang, M. Ishii, Two-phase interfacial structure of bubbly-to-slug transition flows in a 12.7 mm ID vertical tube, Int. J. Heat. Mass Transf. 165 (2021) 120556.
[10] T. Hibiki, P. Ju, S. Rassame, S. Miwa, X. Shen, T. Ozaki, Channel size effect on drift-flux parameters for adiabatic and boiling two-phase flows, Int. J. Heat Mass Transf. 185 (2022) 122410.
[11] T. Hibiki, M. Ishii, One-dimensional drift–flux model for two-phase flow in a large diameter pipe, Int. J. Heat. Mass Transf. 46 (10) (2003) 1773–1790.
[12] S. Rassame, T. Hibiki, Drift-flux correlation for gas-liquid two-phase flow in a horizontal pipe, Int. J. Heat. Fluid. Flow. 69 (2018) 33–42.
[13] T. Hibiki, N. Tsukamoto, Drift-flux model for upward dispersed two-phase flows in vertical medium-to-large round tubes, Progr. Nucl. Energy 158 (2023) 104611.
[14] H. Goda, T. Hibiki, S. Kim, M. Ishii, J. Uhle, Drift-flux model for downward two-phase flow, Int. J. Heat. Mass Transf. 46 (25) (2003) 4835–4844.
[15] C. Dong, T. Hibiki, Drift-flux parameter modeling of vertical downward gas–liquid two-phase flows for interfacial drag force formulation, Nucl. Eng. Des. 378 (2021) 111185.
[16] S.M. Bhagwat, A.J. Ghajar, A flow pattern independent drift flux model based void fraction correlation for a wide range of gas–liquid two phase flow, Int. J. Multiph. Flow 59 (2014) 186–205.
[17] M.A. Woldesemayat, A.J. Ghajar, Comparison of void fraction correlations for different flow patterns in horizontal and upward inclined pipes, Int. J. Multiph. Flow 33 (4) (2007) 347–370.
[18] T. Hibiki, N. Tsukamoto, Drift-flux model for upward dispersed two-phase flows in a vertical rod bundle, Appl. Therm. Eng. 226 (2023) 120323.
[19] T. Abbs, T. Hibiki, One-dimensional drift-flux correlation for vertical upward two-phase flow in a large size rectangular channel, Progr. Nucl. Energy 110 (2019) 311–324.
[20] S. Rassame, T. Hibiki, Drift-flux model for dispersed adiabatic and boiling two-phase flows in rectangular channels, Int. J. Heat. Mass Transf. 224 (2024) 125270.
[21] N. Wang, B. Sun, P. Gong, Z. Wang, Improved void fraction correlation for two-phase flow in large-diameter annuli, Chem. Eng. Technol. 40 (4) (2017) 745–754.
[22] H. Barati, T. Hibiki, Two-group drift-flux model for dispersed gas-liquid flows in annuli, Int. J. Heat. Mass Transf. 225 (2024) 125448.
[23] M. Raissi, P. Perdikaris, G.E. Karniadakis, Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations, J. Comput. Phys. 378 (2019) 686–707.
[24] Z. Dang, Improving the one-dimensional interfacial area transport equation using a physics-informed machine learning method, Int. J. Heat. Mass Transf. 201 (2023) 123608.
[25] Z. Dang, M. Ishii, Towards stochastic modeling for two-phase flow interfacial area predictions: A physics-informed reinforcement learning approach, Int. J. Heat. Mass Transf. 192 (2022) 122919.
[26] S. Kim, Interfacial Area Transport Equation and Measurement of Local Interfacial Characteristics, Purdue University, USA, 1999. Purdue University. Ph. D. Thesis.
[27] X. Sun, *Two-Group Interfacial Area Transport Equation for a Confined Test Section* Purdue University, Purdue University, USA, 2001. Ph. D. Thesis.
[28] Y. Liu, Three-Dimensional Interfacial Area Transport in Gas-Dispersed Two-Phase Flow up to Churn-Annular Flow Transition Purdue University, Purdue University, USA, 2008. Ph. D. Thesis.
[29] J.F. Marchaterre, The Effect of Pressure on Boiling Density in Multiple Rectangular Channels, 5522, Argonne National Laboratory, 1956.
[30] H. Zhang, Y. Xiao, H. Gu, D. Liu, Study on bubbly and cap-bubbly flow in a square channel using dual wire-mesh sensors, Int. J. Multiph. Flow 133 (2020) 103461.
[31] V.S. Chalgeri, J.H. Jeong, Flow regime identification and classification based on void fraction and differential pressure of vertical two-phase flow in rectangular channel, Int. J. Heat. Mass Transf. 132 (2019) 802–816.
[32] X. Yang, J.P. Schlegel, Y. Liu, S. Paranjape, T. Hibiki, M. Ishii, Measurement and modeling of two-phase flow parameters in scaled 8× 8 BWR rod bundle, Int. J. Heat. Fluid. Flow. 34 (2012) 85–97.
[33] S. Morooka, T. Ishizuka, M. Iizuka, K. Yoshimura, Experimental study on void fraction in a simulated BWR fuel assembly (evaluation of cross-sectional averaged void fraction), Nucl. Eng. Des. 114 (1) (1989) 91–98.
[34] T.M. Anklam, R.F. Miller, Void fraction under high pressure, low flow conditions in rod bundle geometry, Nucl. Eng. Des. 75 (1) (1983) 99–108.
[35] H. Kumamaru, M. Kondo, H. Murata, Y. Kukita, Void-fraction distribution under high-pressure boil-off conditions in rod bundle geometry, Nucl. Eng. Des. 150 (1) (1994) 95–105.
[36] M. Sadatomi, Y. Sato, S. Saruwatari, Two-phase flow in vertical noncircular channels, Int. J. Multiph. Flow 8 (6) (1982) 641–655.
[37] N. Wang, B. Sun, P. Gong, Z. Wang, Improved void fraction correlation for two-phase flow in large-diameter annuli, Chem. Eng. Technol. 40 (4) (2017) 745–754.
[38] B. Ozar, J.J. Jeong, A. Dixit, J.E. Juliá, T. Hibiki, M. Ishii, Flow structure of gas–liquid two-phase flow in an annulus, Chem. Eng. Sci. 63 (15) (2008) 3998–4011.
[39] E.F. Caetano, O. Shoham, J.P. Brill, Upward vertical two-phase flow through an annulus—Part II: modeling bubble, slug, and annular flow, J. Energy Resour. Technol. 114 (1) (1992) 14–30. Mar 1992.
[40] Z. Dang, Z. Yang, X. Yang, M. Ishii, Experimental study of vertical and horizontal two-phase pipe flow through double 90 elbows, Int. J. Heat. Mass Transf. 120 (2018) 861–869.

[41] Z. Dang, Z. Yang, X. Yang, M. Ishii, Experimental study on void fraction, pressure drop and flow regime analysis in a large ID piping system, Int. J. Multiph. Flow 111 (2019) 31–41.
[42] K. Usui, K. Sato, Vertically downward two-phase flow,(I) void distribution and average void fraction, J. Nucl. Sci. Technol. 26 (7) (1989) 670–680.
[43] R. Kong, Characterization of Horizontal Air-Water Two-Phase Flow in Different Pipe Sizes, Purdue University, 2018. Doctoral dissertation.
[44] G.H. Abdul-Majeed, Liquid holdup in horizontal two-phase gas—liquid flow, J. Petrol. Sci. Eng. 15 (2-4) (1996) 271–280.
[45] ... & Z. Dang, G. Wang, P. Ju, X. Yang, R. Bean, M. Ishii, M. Bernard, Experimental study of interfacial characteristics of vertical upward air-water two-phase flow in 25.4 mm ID round pipe, Int. J. Heat. Mass Transf. 108 (2017) 1825–1838.
[46] Z. Dang, Experimental Study of Two Phase Upward Flow in Vertical One-Inch Pipe, Purdue University, 2016. Master's thesis.
[47] G. Wang, M. Zhang, Z. Dang, M. Ishii, Axial interfacial area transport and flow structure development in vertical upward bubbly and slug flow, Int. J. Heat. Mass Transf. 169 (2021) 120919.